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LAB_03

Aim: To design and represent an Entity Relationship (ER) diagram for a real-world database system using the draw.io tool by identifying entities, attributes, keys, relationships and cardinalities.

Integrated University Research Management System:

Design a complete ER model for a University Research Management System. The system should maintain information about Researchers, Departments, Research Projects, Students, Publications, and Project Dependents/Assistants.

A Researcher has a unique Researcher_ID, Name, Date_of_Birth, Email, multiple Phone Numbers, and Address. Address consists of Street, City, State, and PIN Code. Age is calculated from Date_of_Birth.

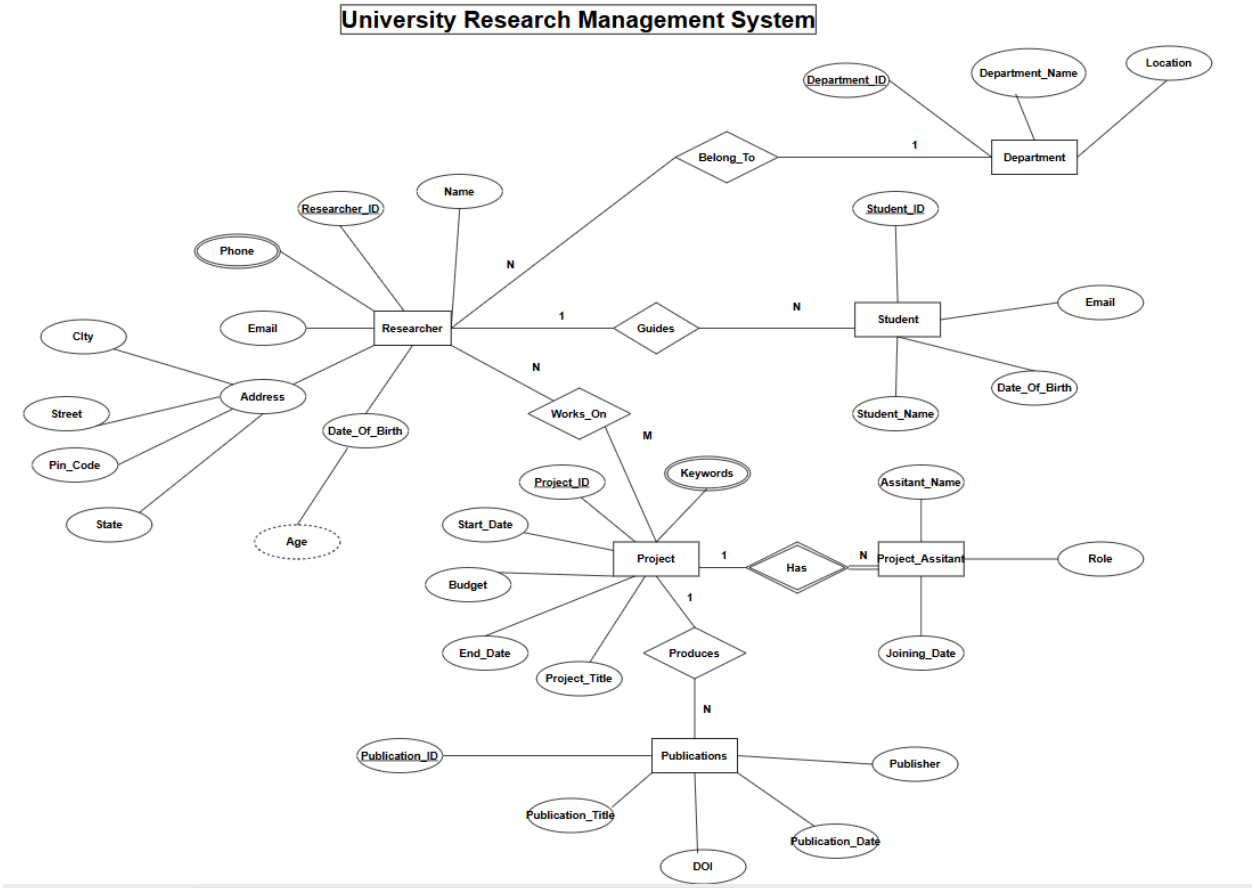
Each researcher belongs to a Department. A department has Department_ID, Department_Name, and Location.

A researcher can work on multiple Research Projects. Each project has Project_ID, Project_Title, Start_Date, End_Date, Budget, and multiple Keywords. A project may involve multiple researchers.

A researcher can have multiple Students working under them. Each student has Student_ID, Name, Email, and Date_of_Birth.

Each research project can produce multiple Publications. A publication has Publication_ID, Title, Publication_Date, DOI, and Publisher.

A project may also have Project Assistants. An assistant is identified only within the context of its project and has Assistant_Name, Role, and Joining_Date. The same assistant name may occur in different projects.



Tasks:

1. Identify all entity types and entity sets.

There are two types of entities: strong entity and weak entity. Strong entities are Researcher, Department, Project, Student and Publication. Project Assistant is a weak entity.

2. Identify strong and weak entity sets.

Strong Entity Sets: Researcher, Department, Project, Student, Publication.
Weak Entity Set: Project Assistant.

3. Identify the owner entity and partial key of the weak entity.

The owner entity is Project. The partial key of Project Assistant is Assistant_Name. Its complete identification is Project_ID + Assistant_Name.

4. Identify simple attributes.

Researcher_ID, Name, Date_of_Birth, Email, Phone Number, Street, City, State, PIN Code, Department_ID, Department_Name, Location, Project_ID, Project_Title, Start_Date, End_Date, Budget, Keywords, Student_ID, Student_Name, Student Email, Student Date_of_Birth, Publication_ID, Title, Publication_Date, DOI, Publisher, Assistant_Name, Role and Joining_Date.

5. Identify composite attributes.

Address is a composite attribute. It consists of Street, City, State and PIN Code.

6. Identify single-valued attributes.

Researcher_ID, Name, Date_of_Birth, Email, Address, Department_ID, Department_Name, Location, Project_ID, Project_Title, Start_Date, End_Date, Budget, Student_ID, Student_Name, Student Email, Student Date_of_Birth, Publication_ID, Title, Publication_Date, DOI, Publisher, Assistant_Name, Role and Joining_Date.

7. Identify multi-valued attributes.

Phone Number and Keywords are multi-valued attributes. A researcher can have multiple phone numbers and a project can have multiple keywords.

8. Identify stored attributes.

All attributes except Age are stored attributes. Age is calculated from Date_of_Birth.

9. Identify derived attributes.

Age is a derived attribute because it is calculated from Date_of_Birth.

10. Identify key attributes.

Researcher_ID, Department_ID, Project_ID, Student_ID and Publication_ID are primary key attributes. Assistant_Name is the partial key of Project Assistant.

11. Identify candidate keys and select appropriate primary keys.

Researcher: Researcher_ID (PK)
Department: Department_ID (PK)
Project: Project_ID (PK)
Student: Student_ID (PK)
Publication: Publication_ID (PK)
Project Assistant: Project_ID + Assistant_Name (composite primary key).

12. Identify at least one composite key.

Project Assistant has the composite key (Project_ID, Assistant_Name). The Works_On relation also has the composite key (Researcher_ID, Project_ID).

13. Identify appropriate foreign keys after converting the ER model into relations.

Researcher.Department_ID → Department(Department_ID)
Student.Researcher_ID → Researcher(Researcher_ID)
Publication.Project_ID → Project(Project_ID)
Project_Assistant.Project_ID → Project(Project_ID)
Works_On.Researcher_ID → Researcher(Researcher_ID)
Works_On.Project_ID → Project(Project_ID).

14. Identify all major relationships between the entities.

Belongs_To: Researcher–Department (N:1)
Works_On: Researcher–Project (M:N)
Guides: Researcher–Student (1:N)
Produces: Project–Publication (1:N)
Has: Project–Project Assistant (1:N).

15. Draw the complete ER diagram using standard ER notation.

The complete ER diagram is prepared in draw.io using standard ER notation. It includes entities, relationships, keys, cardinalities, composite, multi-valued, derived and weak-entity notation.

16. Prepare a table showing each attribute and its attribute type and key status.

Entity	Attribute	Attribute Type	Key Status
Researcher	Researcher_ID	Simple, Single-valued	Primary Key
Researcher	Name	Simple, Single-valued	Non-key
Researcher	Date_of_Birth	Simple, Stored	Non-key
Researcher	Age	Derived	Non-key
Researcher	Email	Simple, Single-valued	Non-key
Researcher	Phone Number	Multi-valued	Non-key
Researcher	Address	Composite	Non-key
Address	Street, City, State, PIN Code	Simple	Non-key
Department	Department_ID	Simple, Single-valued	Primary Key
Department	Department_Name	Simple, Single-valued	Non-key
Department	Location	Simple, Single-valued	Non-key
Project	Project_ID	Simple, Single-valued	Primary Key
Project	Project_Title	Simple, Single-valued	Non-key
Project	Start_Date	Simple, Single-valued	Non-key
Project	End_Date	Simple, Single-valued	Non-key
Project	Budget	Simple, Single-valued	Non-key
Project	Keywords	Multi-valued	Non-key
Student	Student_ID	Simple, Single-valued	Primary Key
Student	Student_Name	Simple, Single-valued	Non-key
Student	Email	Simple, Single-valued	Non-key
Student	Date_of_Birth	Simple, Single-valued	Non-key

Publication	Publication_ID	Simple, Single-valued	Primary Key
Publication	Title	Simple, Single-valued	Non-key
Publication	Publication_Date	Simple, Single-valued	Non-key
Publication	DOI	Simple, Single-valued	Non-key
Publication	Publisher	Simple, Single-valued	Non-key
Project Assistant	Assistant_Name	Simple	Partial Key
Project Assistant	Role	Simple, Single-valued	Non-key
Project Assistant	Joining_Date	Simple, Single-valued	Non-key